

Executive Summary: Coffee Shop Strategic Sales Analysis

1. Executive Overview

The objective of this initiative was to engineer a robust business intelligence framework by transforming 149,116 rows of raw transactional data into high-level strategic assets. As Senior Business Intelligence Lead, I directed the development of a PostgreSQL-driven analytical layer designed to normalize transaction volume and mitigate operational friction.

The technical methodology prioritized query modularity and scalability through:

- **Common Table Expressions (CTEs):** Utilized to construct modular data layers, such as `monthly_sales` and `sales_with_mom`, ensuring complex multi-stage aggregations remained performant and auditable.
- **Window Functions:** Specifically `LAG(...)` `OVER (ORDER BY month)`, used to derive precise Month-over-Month (MoM) growth trajectories for sales, orders, and unit volume.
- **Time-Series Normalization:** Leveraging `DATE_TRUNC` and `EXTRACT` functions to parse temporal patterns, enabling the transition from raw timestamps to strategic day-part and day-type intelligence.
- **Aggregate Logic:** Employed to calculate store-level benchmarks and product-category dominance, providing the empirical foundation for resource reallocation.

2. Growth & Momentum Analytics (Jan–Jun)

The first half of the year demonstrated significant upward momentum. Most notably, the analysis recorded a **31.77% Month-over-Month (MoM) growth in Total Sales during the month of May**. This peak serves as the baseline for our current "Demand Forecasting & Churn Mitigation" strategy, where SQL-driven status flags (e.g., "BELOW AVERAGE") trigger automated customer engagement before churn occurs.

Metric	Business Signal	Strategic Recommendation
Total Sales MoM	Momentum & Revenue Velocity	Anchoring expansion budgets in high-velocity months like May.
Order Count Volatility	Operational Throughput Demand	Direct indicator for scheduling part-time shifts during transaction surges.
Unit Volume Trends	Supply Chain Requirements	Normalizing inventory orders to prevent stock-outs during high-qty periods.

3. Operational Excellence & Resource Allocation

Our "Location Performance Benchmarking" identifies **Hell's Kitchen** as a primary performance anchor, setting the standard for regional logistics. By applying granular timing analysis, we have pinpointed a critical service window between **08:00 AM and 10:00 AM**.

- **Service Peak Analysis:** Data confirms that sales peak at **£18,822.31** during the 08:00 hour and reach a maximum of **£19,639.13** at the 10:00 hour.
- **Throughput Intelligence:** While sales are high at 08:00, the peak **Order Count (4,155)** occurs at hour 10. This indicates that while the early hour may involve higher-ticket items, the 10:00 window requires maximum throughput speed and "all-hands" staffing to maintain service quality.
- **Actionable Move:** Staffing levels are now synchronized with these sub-hour surges to eliminate service bottlenecks and preserve customer experience (CX) during high-pressure intervals.

4. Revenue & Product Portfolio Optimisation

Using May's performance data, we have identified a clear hierarchy within the product mix to drive margin expansion.

- **Category Hierarchy:** **Coffee** remains the undisputed revenue leader at **£60,362.85**, followed by **Tea** at **£44,539.85**. Notably, the **Bakery** category follows at **£18,565.52**, representing a significant drop-off that identifies a potential opportunity for cross-sell optimization.
- **Hero Products:** The **Barista Espresso** is our leading individual product type, generating **£20,423.75**, followed closely by **Brewed Chai Tea** at **£17,427.35**.

- **Pricing Intelligence:** Utilizing `EXTRACT(DOW FROM transaction_date)`, we have differentiated spending behaviors between weekends (0,6) and weekdays. This enables **dynamic weekend surge pricing** on premium categories and **weekday morning loyalty incentives** to maintain high volume during the commuter rush.

5. Strategic Outcomes & ROI Scaling

The implementation of this PostgreSQL analytical framework delivers four definitive business outcomes:

- **Higher Retention:** Proactive identification of downward sales status allows for early intervention in top-tier stores.
- **Cost Reduction:** Precise alignment of staffing and inventory to the 08:00–10:00 surge window significantly reduces labor waste and spoilage.
- **Margin Expansion:** Strategic "doubling down" on high-margin "Hero Products" like **Barista Espresso** and **Brewed Chai Tea** anchors the product mix for maximum profitability.
- **Enhanced Customer Experience (CX):** Data-backed support for high-pressure hours ensures that location-specific friction is mitigated before it impacts brand reputation.